



AYE TECHENGINE | COURSE GUIDE

Fundamentals of Electrical Engineering

Chapters 1 & 2 — Signals, Systems & Complex Numbers

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*Re-typeset from "Fundamentals of Electrical Engineering I"
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About This Guide

This **AYE TechEngine** course guide re-typesets **Chapters 1 and 2** of *Fundamentals of Electrical Engineering I*, the open textbook from the Open University of Hong Kong, into a clean, branded, study-ready format. It introduces the core ideas every electrical engineer carries for life: **what a signal is, what a system is, and the mathematics of sinusoids and complex numbers**.

We have re-organised the content, added our own callouts and worked summaries, and recreated the diagrams in the AYE Tech Hub navy & cyan style. The underlying material is used under the Creative Commons Attribution-ShareAlike 4.0 International License (<http://creativecommons.org/licenses/by-sa/4.0/>). You are free to share and adapt this guide under the same terms.

■ Who This Guide Is For

Engineering students, technicians, and self-learners on the AYE TechEngine track. We assume comfort with high-school algebra and trigonometry; the new material — complex numbers, signals, block diagrams — is introduced from scratch.

■ Learning Outcomes

After completing this guide you will be able to: **(a)** distinguish analog and digital signals; **(b)** draw the five-block model of a communication system; **(c)** manipulate complex numbers in rectangular and polar form; **(d)** recognise the elemental signals (sinusoid, complex exponential, unit step, unit sample, pulse, square wave); **(e)** identify whether a system is linear and time-invariant.

Chapter 1 — Introduction

1.1 Themes

Electrical engineering grew up around two great themes: **power** (creating, transmitting and consuming electrical energy) and **information** (representing, manipulating, transmitting and receiving messages electrically). This course concentrates on the second theme — what information is, how engineers quantify it, and how electrical signals can carry it.

The conversion of an information-bearing signal from one energy form into another (acoustic to electrical, electrical to optical, and so on) is called **energy conversion** or **transduction**. Every conversion is imperfect — some energy is always lost as heat — but the *information* is not necessarily lost.

We will be concerned with how to **represent** all forms of information using electrical signals, **encode** that information as voltages, currents, and electromagnetic waves, **transmit** it (over wires, fibres or free space), **receive** it cleanly, and ultimately **decode** it back into a usable form.

1.2 Signals Represent Information

Whether the information source is speech, music, video, or sensor data, a **signal** is the physical quantity (typically a voltage or current) we use to represent it. The independent

variable of the signal can be time (speech), space (images), or the integers (a stream of symbols). Engineers categorise information into two forms:

- **Analog** — continuous-valued; examples include audio waveforms, video brightness, and most sensor outputs.
- **Digital** — discrete-valued; examples include text, DNA sequences, and digitised audio and video.

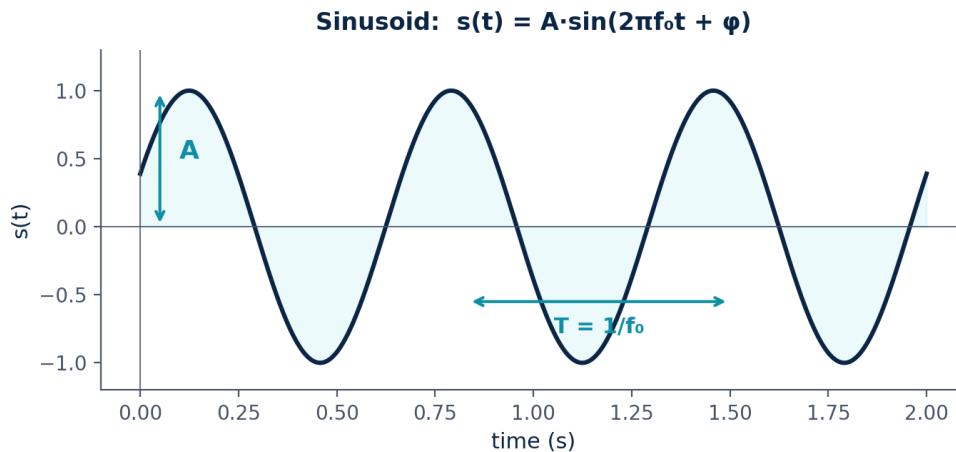


Figure 1 — An analog signal: amplitude varies continuously with time. Speech, music and most sensor outputs look like this.

■ Why Analog Becomes Digital

Digital signals are vastly easier to store, copy, encrypt, and transmit without loss. That is why nearly every modern communication system — telephone, radio, television, internet — first samples the analog source, converts it to digital using an **analog-to-digital converter (ADC)**, processes the bits, and finally regenerates an analog output for human senses through a **digital-to-analog converter (DAC)**.

1.3 Structure of Communication Systems

Every communication system — from a baby monitor to a deep-space probe — fits the same five-block model. A **source** emits a message $s(t)$. A **transmitter** modulates it into a form $x(t)$ that can travel through a physical **channel**. The channel adds noise and distortion, producing a corrupted signal $r(t)$. A **receiver** demodulates and cleans this back into an estimate $\hat{s}(t)$ of the original, which is delivered to a **sink** (a loudspeaker, a screen, a human, or another system).

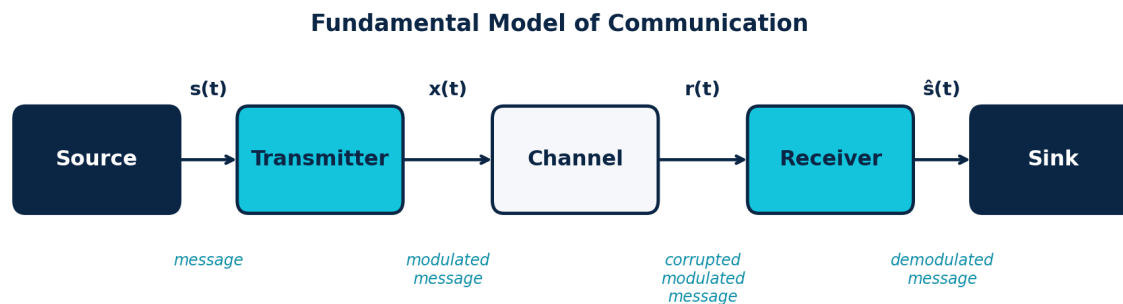


Figure 2 — The fundamental five-block model of a communication system (source → transmitter → channel → receiver → sink).

More generally, in electrical engineering we draw any process as a **system** — a block that takes input signals and produces output signals. The block diagram below is the most basic system, with one input $x(t)$ and one output $y(t)$:

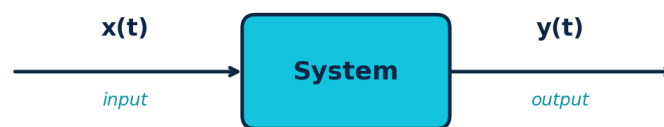


Figure 3 — A system operates on its input signal $x(t)$ to produce an output signal $y(t)$. The notation is identical whether the system is an amplifier, a filter, or an entire data-centre.

1.4 The Fundamental Signal — Sinusoids

The most important signal in all of electrical engineering is the **sinusoid**:

$$s(t) = A \sin(2\pi f_0 t + \varphi)$$

It is described by three parameters: **amplitude** A (the peak height), **frequency** f_0 in hertz (cycles per second), and **phase** φ in radians (the horizontal offset). The **period** is $T = 1/f_0$. The angular frequency is $\omega = 2\pi f_0$ in radians per second.

■ Why Sinusoids Are Special

Sinusoids are the only signals whose *shape* is preserved by every linear, time-invariant system — amplifiers, filters, transmission lines, even loudspeakers. The output may have a different amplitude or phase, but it is still a sinusoid at the same frequency. This single property is the reason Fourier analysis underpins almost the entire field of signal processing.

1.5 Communicating Information with Signals

A pure sinusoid carries one bit of information: "I am present." To transmit anything more — speech, music, data — we must **modulate** one or more of its parameters. Three classical schemes (still used today, in evolved form, by every radio, modem and cell-phone) are:

- **Amplitude modulation (AM)** — vary the amplitude A with the message.
- **Frequency modulation (FM)** — vary the frequency f_0 with the message.
- **Phase modulation (PM)** — vary the phase ϕ with the message.

Modern digital systems combine these with bit-coding and pulse shaping to produce sophisticated schemes like QAM, OFDM and CDMA — but the underlying carrier is still a sinusoid.

Chapter 2 — Signals and Systems

Sinusoids are at the heart of signal analysis, but to manipulate them mathematically we need **complex numbers**. Chapter 2 develops the algebra of complex numbers and uses it to build a library of **elemental signals** — the LEGO bricks from which every more complex signal is assembled.

2.1 Complex Numbers

A **complex number** has a real part and an imaginary part. In engineering we write the imaginary unit as j (electrical engineers reserve i for current):

$$z = a + jb = r e^{j\theta}, \quad r = |z| = \sqrt{a^2 + b^2}$$

The **complex plane** is a coordinate system whose horizontal axis is the real part and vertical axis is the imaginary part. A complex number can be expressed in two equivalent forms:

- **Rectangular form:** $z = a + jb$, where $a = \text{Re}(z)$ and $b = \text{Im}(z)$.
- **Polar form:** $z = r \cdot e^{j\theta}$, where $r = |z| = \sqrt{a^2 + b^2}$ is the magnitude and $\theta = \arctan(b/a)$ is the angle.

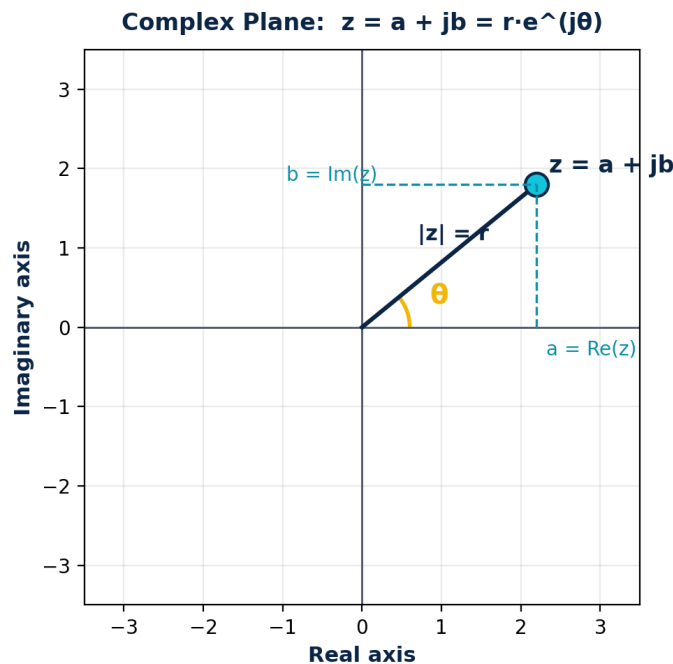


Figure 4 — A complex number $z = a + jb$ plotted in the complex plane. Its magnitude r and angle θ give the polar form $z = r \cdot e^{j\theta}$.

Arithmetic with complex numbers

Addition and subtraction are easiest in rectangular form; multiplication and division are easiest in polar form.

$$z_1 + z_2 = (a_1 + a_2) + j(b_1 + b_2)$$

$$z_1 \cdot z_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$$

$$\bar{z} = a - jb, \quad z\bar{z} = |z|^2 = a^2 + b^2$$

Field Tip — Switch Forms Freely

The polar form makes multiplication trivial (multiply magnitudes, add angles), while the rectangular form makes addition trivial. Skilled engineers switch forms mid-calculation without hesitation. On a calculator, the **rect→polar** button is your best friend.

Euler's relation and the complex exponential

Euler's relation connects the exponential, sine and cosine functions:

$$e^{j\theta} = \cos \theta + j\sin \theta$$

From it we get the two most useful decompositions in signal processing:

$$\cos(2\pi ft) = \frac{e^{j2\pi ft} + e^{-j2\pi ft}}{2}$$

$$\sin(2\pi ft) = \frac{e^{j2\pi ft} - e^{-j2\pi ft}}{2j}$$

The **complex exponential** is the single most important signal in all of electrical engineering:

$$s(t) = Ae^{j(2\pi f_0 t + \varphi)} = Ae^{j\varphi} e^{j2\pi f_0 t}$$

Its magnitude is A , its phase is $2\pi f_0 t + \varphi$, and it rotates around the origin of the complex plane at a constant rate. We can recover real-valued sinusoids by taking real or imaginary parts:

$$A\cos(2\pi ft + \varphi) = \text{Re}(Ae^{j\varphi} e^{j2\pi ft})$$

2.2 Elemental Signals

Engineers build a library of **elemental signals** to use as building blocks. Mastering these is essential.

Signal	Continuous Form	Use
Sinusoid	$A\sin(2\pi f_0 t + \varphi)$	Carriers, AC sources, vibrations
Complex exponential	$Ae^{j(2\pi f_0 t + \varphi)}$	Phasor analysis, Fourier transforms
Real exponential	$Ae^{-t/\tau}$	RC/RL transients, decay processes
Unit step $u(t)$	0 for $t < 0$, 1 for $t \geq 0$	"Switch-on" events
Pulse $p(t)$	1 on an interval, 0 elsewhere	Digital bits, gating
Square wave $sq(t)$	Alternating $\pm A$ periodically	Clocks, PWM

Two signals deserve special attention: the **unit step** (continuous time) and the **unit sample** or **impulse** (discrete time). They are the building blocks for representing switches and

sampled inputs:

$$u(t) = 1 \text{ for } t \geq 0; \quad u(t) = 0 \text{ for } t < 0$$

$$\delta[n] = 1 \text{ for } n = 0; \quad \delta[n] = 0 \text{ otherwise}$$

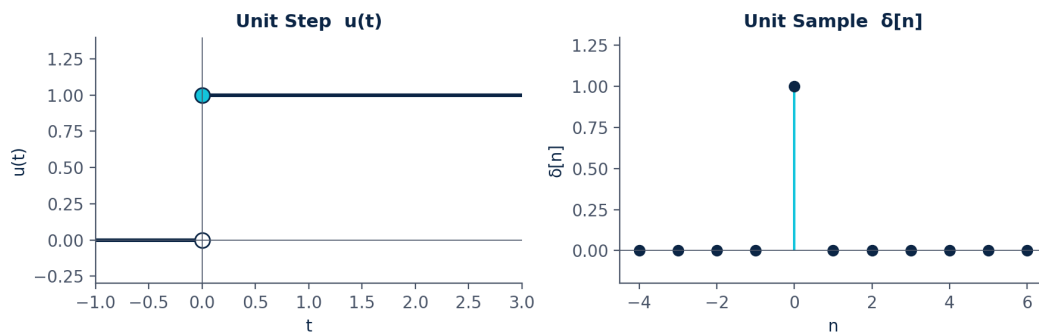


Figure 5 — Left: the continuous-time unit step $u(t)$. Right: the discrete-time unit sample $\delta[n]$.

2.3 Discrete-Time Signals

When an analog signal is sampled by an ADC, the result is a **discrete-time signal** — a sequence of values $s[n]$ indexed by integers $n = \dots, -2, -1, 0, 1, 2, \dots$. The elemental signals reappear in discrete-time form:

- **Discrete sinusoid:** $s[n] = A \cdot \sin(2\pi f_0 n + \phi)$.
- **Discrete complex exponential:** $s[n] = e^{(j2\pi f_0 n)}$.
- **Unit sample:** $\delta[n] = 1$ only at $n = 0$, zero elsewhere.
- **Symbolic-valued signals:** each $s[n]$ takes values from a finite alphabet — the natural representation of text and other purely digital sources.

■ Signals vs. Systems

Be precise with vocabulary: a **signal** is data (a function of time, space, or integers). A **system** is a process — it *operates on* signals. An amplifier is a system; the voltage out of it is a signal.

2.4 Simple Systems

Most useful systems are built by interconnecting simpler systems. Three topologies cover almost every case:

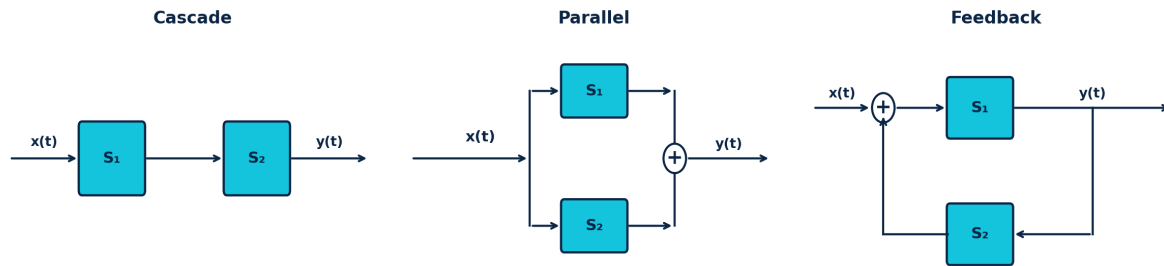


Figure 6 — The three fundamental system interconnections. Cascade: S_1 then S_2 . Parallel: S_1 and S_2 act on the same input and their outputs sum. Feedback: the output is fed back, subtracted (or added) from the input.

Five elementary systems are worth memorising:

System	What It Does	Equation
Source	Produces a signal regardless of input	$y(t) = \text{some signal}$
Amplifier	Scales the input by a constant gain G	$y(t) = G \cdot x(t)$
Delay	Shifts the input later in time	$y(t) = x(t - \tau)$
Time-reversal	Flips the input in time	$y(t) = x(-t)$
Integrator / Derivative	Integrates or differentiates input	$y(t) = \int x(t)dt$ or $y(t) = dx/dt$

Two further properties decide the entire mathematical treatment of a system:

- **Linearity** — a system is linear if its response to a sum of scaled inputs equals the same sum of scaled responses. Formally: if $x_1(t) \rightarrow y_1(t)$ and $x_2(t) \rightarrow y_2(t)$, then $a \cdot x_1(t) + b \cdot x_2(t) \rightarrow a \cdot y_1(t) + b \cdot y_2(t)$.
- **Time-invariance** — a system is time-invariant if shifting the input in time produces the same shift in the output. If $x(t) \rightarrow y(t)$, then $x(t - \tau) \rightarrow y(t - \tau)$.

■ LTI: The Goldilocks Class

A system that is both **Linear and Time-Invariant (LTI)** can be fully described by a single function — its **impulse response** $h(t)$ — and analysed exclusively with sinusoids and complex exponentials. This is why LTI systems dominate undergraduate courses, professional design, and the rest of this book.

Summary & Key Takeaways

Concept	Essential Point
Signal	A function that carries information
Analog vs Digital	Continuous vs discrete in value

Comm. system	Source → Transmitter → Channel → Receiver → Sink
Sinusoid	$A \cdot \sin(2\pi f_0 t + \varphi)$ — the universal carrier
Complex number	$z = a + jb = r \cdot e^{j\theta}$
Euler's relation	$e^{j\theta} = \cos\theta + j \cdot \sin\theta$
Complex exponential	The most important signal — phasor analysis hinges on it
System	Operates on signals to produce signals
LTI system	Linear & Time-Invariant — fully described by its impulse response

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